

First Bug:- SQL search filter problem

where I found bug:-

Database/queries/search_tasks.sql

Database/Oracle/task-search-package.sql.

What problem?

I noticed that both using AND and OR without parentheses.

Previous:-

WHERE archived=FALSE

AND LOWER(title) LIKE :term

OR LOWER(description) LIKE :term

AND (:status IS NULL OR status=:status)

SQL gives higher priority to AND than OR.

So database does not read, one combined search condition

I Fixed:-

by added parentheses around the title and description

WHERE archived=FALSE

AND (

LOWER(title) LIKE :term

OR LOWER(description) LIKE :term

)
AND (:status IS NULL OR status=:status)

Why we fix the problem

because now the database first checks the search term is present in either title or description
* it applies archived = FALSE condition and the Status Filter.

also same correction in Oracle reference

Bug 2:-

The API accepts invalid values for:

Page, pageSize

example:-

api/fake?

Page = 0

Page = -1

Page Size = 0

Why is bug:-

pagination calculation depends on these values

Added validation checks:

1) if (page < 1)

2) if (pageSize < 1 || pageSize > 100)

and return: HTTP 400 Bad Request

* it ensures pagination behavior.

* prevents invalid request.

Bug 3:-

Search / Filter pagination Reset

problems-

- * user changes the search text or status filter
- * the application may request a page that no longer exists for the filtered results.

view:- App.jsx and verified.

set page (1);

- * Search text changes
- * Status filter changes

It resolved in the current implementation.

Bug 4:- Stale search Results.

problem :-

- * Multiple search requests may be sent while the user types quickly.

Impact -

An older request may finish after a new request and override correct results

Javascript:- Use Tasks.js

Abort Controller

use Tasks.js :-

abortRef.current.abort();

and

and:-

```
new AbortController();  
const controller = new AbortController();
```

Fixed:- stale responses are prevented.

example:-

```
if (abortRef.current) {  
  abortRef.current.abort();  
}
```

Result:-

old requests are cancelled

Last request work

Search result remain accurate.